

Cardio AI Clinical Copilot - Project Report

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Overview

This project is a full-stack clinical decision-support environment combining realtime voice transcription, symptom extraction, AI

Core Workflow

1. Doctor starts consultation workflow.
2. Patient speech or manual transcript is captured.
3. Input language is selected; transcript is normalized to English for downstream reasoning.
4. Symptoms are extracted and mapped to risk and body regions.
5. AI recommendations update in realtime: doctor questions, tests, diagnosis direction, and next steps.
6. Doctor can upload report images for AI analysis.
7. Disease-focused notes are accumulated.
8. System generates and exports a consultation PDF.

How Agents Work Together

- nlp_symptom_agent: symptom extraction (English + Urdu/Roman Urdu support).
- feature_mapper_agent: symptom-to-feature mapping.
- prediction_agent: model inference and probability.
- explainability_agent: SHAP-based reasoning.
- doctor_copilot_agent: deterministic fallback question/test guidance.
- pain_mapper_agent + body map logic: pain localization.
- heart_region_agent: cardiac-region hypotheses.
- uncertainty_agent: confidence bucket and score.
- causal_inference_agent: counterfactual hypotheses.
- multimodal_fusion_agent: transcript + report + risk fusion.
- guidelines_agent: protocol-aligned guidance.
- knowledge_graph_agent: clinical entity linking.
- evidence_retrieval_agent: concise evidence snippets.
- fairness_bias_agent: fairness-evaluation snapshot.
- robustness_agent: input quality checks.
- human_feedback_agent: feedback capture stub.
- evaluation_agent: validation/readiness snapshot.

Backend APIs

- POST /diagnose
- POST /ai-insights
- POST /analyze-report-image
- POST /final-report
- GET /research-agents
- WS /ws/consultation

Frontend

- /workflow/start
- /workflow/session
- Home / Diagnose / History

Current Status

The app operates as a workflow-first clinical copilot with realtime AI guidance, multilingual input support, body pain visualizati

Safety Note

This system provides decision support only and does not replace licensed clinical judgment.